

Clinical Study Summary Report (CSSR)

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1. Administrative & Study Identification

Full Study Title: An Open-Label, Randomised, Multi-Drug, Biomarker-Directed, Multi-Centre, Multi-arm Phase 1b Study in Patients With Muscle Invasive Bladder Cancer (MIBC) Who Have Progressed on Prior Treatment (BISCAY) **Short Title/Acronym:** BISCAY Study **Protocol Number:** D4191C00030 **NCT Number:** NCT02546661 **Trial Status:** Active, not recruiting **Sponsor/Company Name:** AstraZeneca **Investigational Product:** Durvalumab (MEDI4736, anti-PD-L1) in combination with targeted therapies (AZD4547, Olaparib [AZD2281, Lynparza], AZD1775, Vistusertib [AZD2014], AZD9150, Selumetinib [AZD6244]) **Phase of Development:** Phase 1b **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability, and frequency/nature of adverse events of multiple targeted agents as monotherapy and in combination with durvalumab (MEDI4736) in biomarker-selected, chemotherapy-refractory advanced urothelial/muscle invasive bladder cancer. **Secondary Objectives:** To evaluate Objective Response Rate (ORR), Disease Control Rate (DCR), Progression-Free Survival (PFS), Overall Survival (OS), Duration of Response (DoR), and evaluate the predictive value of circulating tumor DNA (ctDNA) and other biomarkers.

2.2 Background & Rationale To address the clinical need in advanced urothelial cancer (AUC) and Muscle Invasive Bladder Cancer (MIBC), which is characterized by high mutational burden and recurrent targetable genomic alterations. Despite standard-of-care chemotherapy or immunotherapies, relapse rates remain high. This modular, biomarker-driven "umbrella" study evaluates if matching specific genomic alterations (e.g., FGFR mutations, homologous recombination repair [HRR] deficiency, mTOR/PI3K pathway alterations) with combinations of targeted therapies and immune checkpoint blockade (durvalumab) can overcome resistance and improve response rates compared to monotherapy.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Adaptive, multi-arm "umbrella" design) **Control Type:** Active Comparator/Monotherapy arms (e.g., AZD4547 monotherapy, Durvalumab monotherapy) **Blinding:** Open-label **Randomization:** Randomized (1:1 allocation in specific modules) and Biomarker-directed

3.2 Planned Enrollment & Duration Targeted Enrollment: 117 Number of Sites: Global multi-centre study (e.g., UK, France, Spain, USA, Canada) **Estimated Study Start:** October 2016 **Estimated Primary Completion:** March 2019

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Metastatic Muscle Invasive Bladder Cancer (MIBC) / Advanced Urothelial Cancer. 2. Disease progression following prior adjuvant/neo-adjuvant or 1st-line platinum-based chemotherapy (2nd/3rd line setting; failed adjuvant/neo-adjuvant chemotherapy <1 yr). 3. Identification of eligible genomic alterations (e.g., FGFRm, HRRm) via tumor profiling (for specific biomarker-directed modules). 4. At least 1 measurable lesion ≥ 10 mm at baseline in the longest diameter suitable for accurate repeated measurement per RECIST 1.1. 5. WHO/ECOG performance status 0 or 1.

4.2 Exclusion Criteria 1. Immunotherapy, chemotherapy, anticancer agents, or radiotherapy <4 weeks prior to treatment. 2. Prior exposure to immune checkpoint inhibitors (anti-PD-1, anti-PD-L1) or targeted agents specific to the assigned module. 3. Active autoimmune disease (≤ 2 yr), primary immunodeficiency, or requirement for immunosuppressive drugs within 28 days. 4. Untreated or symptomatic central nervous system (CNS) metastases or spinal cord compression. 5. Clinically significant cardiovascular disease, LVEF $< 55\%$, or mean QTc ≥ 470 ms. 6. Inadequate organ/bone marrow function (e.g., ANC $< 1.5 \times 10^9$ /L, Platelets $< 100 \times 10^9$ /L, severe hepatic impairment). 7. Live attenuated vaccination <30 days.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Durvalumab (MEDI4736) administered intravenously (IV) every 4 weeks, paired with module-specific targeted therapy (e.g., AZD4547 twice daily) dosed continuously or intermittently. **Route of Administration:** Intravenous (Durvalumab/MEDI4736, AZD9150) and Oral (AZD4547, Olaparib/Lynparza, AZD1775, Vistusertib/AZD2014, Selumetinib). **Duration of Treatment:** Continued until objective disease progression, unacceptable toxicity, or patient withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care and symptom management. **Prohibited:** Concurrent chemotherapy, immunotherapy, biologic or hormonal therapy; live attenuated vaccines within 30 days; and strong CYP3A4 inhibitors/inducers for modules containing sensitive targeted agents.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, NGS Biomarker Profiling, Baseline Imaging, Safety Labs
Baseline	Day 0	Module Allocation, Randomization, First Dose Administration
Interim	Every 4 weeks	Safety Assessments, PK/PD Blood Draws, Efficacy Evaluation (Tumor Imaging typically Q8W), ctDNA collection
End of Study	At Progression	Final Vital Signs, Exit Interview, and 90-day Safety/Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints Definition: The frequency and nature of adverse events to determine safety and tolerability, specifically targeting: 1. **Module A:** The frequency and nature of adverse events related to AZD4547 monotherapy. 2. **Module A:** The frequency and nature of adverse events related to the combination of intravenous MEDI4736 (durvalumab) and oral AZD4547. 3. **Module B:** The frequency and nature of adverse events related to the combination of intravenous MEDI4736 (durvalumab) and oral olaparib.

7.2 Secondary & Exploratory Endpoints 1. Objective Response Rate (ORR). 2. Disease Control Rate (DCR). 3. Progression-Free Survival (PFS). 4. Overall Survival (OS) and Duration of Response (DoR). 5. Exploratory correlations of genomic alterations in circulating tumor DNA (ctDNA) and tissue biopsies with clinical outcomes.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Assessed continuously at each clinic visit, at study discontinuation, and 90 days post-treatment. **Serious Adverse Events (SAEs):** Routine 24-hour reporting to the Sponsor and regulatory bodies. **Data Monitoring Committee (DMC):** An independent Safety Review Committee to evaluate Dose Limiting Toxicities (DLTs) and guide adaptive module expansions or closures.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) set for efficacy endpoints; Safety Set for all patients receiving at least one dose of study medication. **Sample Size Justification:** Adaptive design driven by biomarker prevalence; total targeted enrollment is 117 patients to populate targeted modules with enough statistical power to evaluate initial safety signals and

efficacy. **Handling of Missing Data:** Data obtained up until objective progression or the last evaluable assessment is included in the ORR/DCR analysis.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Rigorous onsite and centralized monitoring for data integrity, particularly concerning the adaptive allocation of patients to modules based on central genomics. **Audit Trail:** Robust eCRF locking and query resolution processes across all multi-national centers.

11. Study Identifiers & Governance

Primary ID: NCT02546661 **Registry Number:** NCT02546661 **Sponsor/Lead Organization:** AstraZeneca **Company:** AstraZeneca **Study Title:** BISCAY **Official Regulatory Title:** An Open-Label, Randomised, Multi-Drug, Biomarker-Directed, Multi-Centre, Multi-arm Phase 1b Study in Patients With Muscle Invasive Bladder Cancer (MIBC) Who Have Progressed on Prior Treatment

12. Clinical Framework (Oncology/Bladder Cancer Specific)

Primary Condition: Muscle Invasive Bladder Cancer (MIBC) / Advanced Urothelial Cancer (AUC) **Diagnosis Threshold:** Chemotherapy-refractory disease with specific NGS-identified genomic mutations (e.g., FGFRm, HRRm, PI3K/mTOR alterations) **Medical Methodology:** Biomarker-directed Targeted Therapy + Immunotherapy (PD-L1 inhibition) **Optimization Target:** Maximum tolerated dose, Safety/Tolerability, and optimal Objective Response Rate (ORR)

13. Study Procedures & Methodology

Management Pathway: Adaptive biomarker-directed module allocation **Education Protocol (HCP):** Clinical protocol training for adaptive design and immune-related AE (irAE) management **Education Protocol (Patient):** Informed consent emphasizing genomic tissue testing and multi-drug interventions **Quality Audit Mechanism:** Centralized genomic screening verification and independent radiological review for RECIST 1.1 compliance

14. Observation & Follow-Up Schedule

Total Duration: Until disease progression, unacceptable toxicity, or patient withdrawal **Onsite Visit Cadence:** Weekly/bi-weekly during initial cycles, adjusting per module specific parameters **Remote Monitoring Frequency:** 90-day safety follow-up post-treatment discontinuation **Checklist Review Interval:** Tumor imaging assessments every 8 weeks

15. Sign-off & Approval

Role	Name	Signature	Date		----	----	----	----
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					